

Validated Folded-Gaussian Assembly and Exact Stochastic Repair

Algebraic Midpoint Rows, 8σ Truncation, Haar Compression, and a Perron-Exact Upstream Split

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Abstract

RH-70 proved rigorous terminal Hardy bounds for the frozen matrices emitted by the production pipeline, while RH-71 localized the finite-scale open frontier to the upstream operator/source/observation triple. This paper validates the first part of that frontier: folded-Gaussian matrix and Haar block assembly.

Let u_* be the unique root in $(1, 2)$ of $u^3 - 2u^2 + 2u - 2 = 0$. At positive midpoint nodes x_i, y_j , define

$$g_{ij} = \exp\left(-\frac{(y_j - (1 - u_* x_i^2))^2}{2\sigma^2}\right) + \exp\left(-\frac{(-y_j - (1 - u_* x_i^2))^2}{2\sigma^2}\right).$$

The fully normalized row is $p_{ij} = g_{ij} / \sum_j g_{ij}$. If s_i is the same row restricted to the production support and renormalized, then the exact identity

$$\|p_i - s_i\|_1 = 2 \frac{\sum_{j \notin J_i} g_{ij}}{\sum_j g_{ij}}$$

turns the truncation audit into one positive mass ratio.

For a frozen nonnegative binary64 row f_i , leave every entry except its largest unchanged and set the largest entry equal to one minus the exact dyadic sum of the others. Whenever that replacement is positive, the repaired row $\bar{f}_i$ is exactly stochastic,

$$\|\bar{f}_i - f_i\|_1 = \left| 1 - \sum_j f_{ij} \right|,$$

and the constant vector is an exact Perron right vector. Outward-rounded absolute row and column sums R, C then imply

$$\|P - \bar{F}\|_2 \leq \sqrt{RC}.$$

We also propagate this defect through exact and frozen Haar coarse/detail embeddings by an explicit three-term product bound.

A 192-bit Arb audit covers $\sigma = 0.16, 0.08, 0.04, 0.02, 0.01$ and dimensions 32 through 512. Every support-center floor is interval-stable. The largest full-to-sparse row defect is below 9.58×10^{-17} ; the finest $\|P - \bar{F}\|_2$ upper is 1.495×10^{-14} ; and every Haar coarse/cross assembly defect is below 1.538×10^{-14} . Exact row repairs are below 3.87×10^{-16} and every repaired pivot exceeds 0.0775.

Thus kernel evaluation, support selection, sparse renormalization, Haar assembly, and the Perron right vector are green at the archived scales. The remaining finite-scale spectral gate is the stationary Perron left vector and the parity Riesz pair. This paper does not validate rank-two deflation, prove small-noise uniformity, close Stage A1 or A4, construct a Hilbert–Polya operator, derive a $T \log T$ law or prime-power trace formula, or prove the Riemann Hypothesis.

Keywords: folded Gaussian kernel; interval arithmetic; stochastic repair; Perron vector; Haar compression; validated assembly.

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1 Introduction

The quadratic map

$$f_u(x) = 1 - ux^2$$

at its first band-merging parameter supplies the deterministic geometry used throughout the RH series [3, 5]. The noisy positive-state transfer matrix is obtained by folding a Gaussian kernel, truncating to a finite support, and normalizing every row.

RH-70 began only after that process and after spectral deflation: it embedded the resulting binary64 operator, source, and observation arrays as exact dyadic Arb/Acb inputs [4]. RH-71 therefore identified upstream inclusion as the next finite-scale gate [6].

The upstream calculation contains two conceptually different tasks:

1. evaluate and normalize the folded-Gaussian and Haar blocks;
2. extract the stationary left factor and the parity Riesz pair.

The first is positive scalar arithmetic; the second is a nonnormal spectral problem. This paper separates them. Such separation avoids hiding a spectral conditioning issue inside a large interval matrix constructor.

Contributions

1. We certify the algebraic parameter by a monotone polynomial interval.
2. We derive an exact full-versus-support-renormalized row identity.
3. We construct an exact dyadic stochastic repair of every frozen row.
4. We turn entrywise interval differences into induced two-norm bounds.
5. We propagate the matrix and $2^{-1/2}$ defects through coarse/detail Haar products.
6. We execute the entire assembly audit at the five RH-70 scales.

2 The algebraic folded midpoint matrix

Let

$$q(u) = u^3 - 2u^2 + 2u - 2.$$

The parameter u_* is the first band-merging root.

Proposition 2.1 (Certified algebraic parameter). *The interval*

$$I_* = [1.5436890126920763615708559718017479865252032976510 \\ \pm 2.61 \times 10^{-50}]$$

contains exactly one root of q .

Proof. Outward-rounded evaluation gives a negative upper endpoint for q at the left endpoint of I_* and a positive lower endpoint at the right endpoint. Moreover,

$$q'(I_*) = 3I_*^2 - 4I_* + 2 \subset [2.9741712529504070 \dots \pm 3.40 \times 10^{-49}],$$

whose lower endpoint is positive. The intermediate value theorem and strict monotonicity give existence and uniqueness. $\square$

For dimension n , put

$$x_i = y_i = \frac{i + 1/2}{n}, \quad 0 \leq i < n,$$

and $\mu_i = 1 - u_* x_i^2$. The unnormalized folded weights are

$$g_{ij} = e^{-(y_j - \mu_i)^2 / (2\sigma^2)} + e^{-(-y_j - \mu_i)^2 / (2\sigma^2)}. \quad (1)$$

The common Gaussian prefactor cancels under row normalization.

Define the full midpoint matrix

$$P_{ij} = \frac{g_{ij}}{Z_i}, \quad Z_i = \sum_{j=0}^{n-1} g_{ij}. \quad (2)$$

Thus $P\mathbf{1} = \mathbf{1}$ exactly.

3 Exact support-truncation identity

Let J_i be the support used by the production sparse row. Put

$$Z_{i,J} = \sum_{j \in J_i} g_{ij}, \quad Z_{i,\text{out}} = Z_i - Z_{i,J},$$

and

$$S_{ij} = \begin{cases} g_{ij}/Z_{i,J}, & j \in J_i, \\ 0, & j \notin J_i. \end{cases}$$

Theorem 3.1 (Full/sparse row identity). *For every row with positive weights,*

$$\|P_i - S_i\|_1 = 2 \frac{Z_{i,\text{out}}}{Z_i}. \quad (3)$$

Proof. On the retained support, $Z_{i,J} \leq Z_i$, so

$$\sum_{j \in J_i} \left(\frac{g_{ij}}{Z_{i,J}} - \frac{g_{ij}}{Z_i} \right) = 1 - \frac{Z_{i,J}}{Z_i} = \frac{Z_{i,\text{out}}}{Z_i}.$$

The omitted coordinates contribute the same quantity:

$$\sum_{j \notin J_i} \frac{g_{ij}}{Z_i} = \frac{Z_{i,\text{out}}}{Z_i}.$$

Adding proves (3). $\square$

The production support center is

$$c_i = \left\lfloor n|\mu_i| - \frac{1}{2} \right\rfloor$$

with half-width $\lceil 8\sigma n \rceil + 2$. The interval run verifies that the lower and upper endpoints give the same floor for every row. Therefore support selection itself is stable under the certified algebraic parameter interval.

4 Exact stochastic repair

Binary64 row normalization generally leaves a tiny exact-dyadic row-sum defect. Treating the rounded row as exact does not make that defect vanish [1].

Let $f = (f_0, \dots, f_{n-1})$ be a nonnegative dyadic row and choose an index k with maximal entry. Define

$$\bar{f}_j = f_j \quad (j \neq k), \quad \bar{f}_k = 1 - \sum_{j \neq k} f_j. \quad (4)$$

Theorem 4.1 (Exact dyadic stochastic repair). *If the replacement in (4) is positive, then:*

(i) every $\bar{f}_j$ is a nonnegative dyadic rational and $\sum_j \bar{f}_j = 1$ exactly;

(ii)

$$\|\bar{f} - f\|_1 = \left| 1 - \sum_j f_j \right|; \quad (5)$$

(iii) for the rowwise repaired matrix $\bar{F}$, $\bar{F}\mathbf{1} = \mathbf{1}$ exactly and

$$\|\bar{F} - F\|_\infty = \max_i \left| 1 - \sum_j F_{ij} \right|. \quad (6)$$

Proof. Finite sums and differences of dyadic rationals are dyadic. The definition forces exact row sum one. Only coordinate k changes, by $1 - \sum_j f_j$, giving (5). Taking the maximum row ℓ^1 norm gives (6); exact stochasticity gives the Perron right-vector statement. $\square$

The repair need not be stored as binary64: it is an exact dyadic rational accepted directly by Arb. Its floating display is only a convenience.

5 From entry balls to matrix and Haar bounds

Let $D = P - \bar{F}$. Suppose outward-rounded accumulation gives

$$R \geq \max_i \sum_j |D_{ij}|, \quad C \geq \max_j \sum_i |D_{ij}|.$$

Proposition 5.1 (Induced two-norm enclosure).

$$\|D\|_2 \leq \sqrt{\|D\|_1 \|D\|_\infty} \leq \sqrt{CR}. \quad (7)$$

Proof. This is the standard induced-norm inequality, with the two induced norms bounded by the certified absolute column and row sums. $\square$

Let $E, D_H \in \mathbb{R}^{n \times n/2}$ be the exact Haar coarse and detail embeddings with entries $\pm 2^{-1/2}$, and let $E_0, D_{H,0}$ use the frozen binary64 constant. Their defects satisfy

$$\delta_H = \|E - E_0\|_2 = \|D_H - D_{H,0}\|_2 = \sqrt{2} \left| 2^{-1/2} - \text{fl}(2^{-1/2}) \right|.$$

Theorem 5.2 (Haar compressed assembly bound). *Let X, Y independently denote either the exact coarse or detail embedding, and X_0, Y_0 the matching frozen embeddings. Then*

$$\|X^*PY - X_0^*\bar{F}Y_0\|_2 \leq \|P - \bar{F}\|_2 \delta_H \|\bar{F}\|_2 \|X_0\|_2 \|\bar{F}\|_2 \delta_H. \quad (8)$$

The same upper therefore applies to the coarse matrix and both coarse/detail cross blocks.

Proof. Insert and subtract $X^*\bar{F}Y$ and $X_0^*\bar{F}Y$:

$$X^*PY - X_0^*\bar{F}Y_0 = X^*(P - \bar{F})Y + (X - X_0)^*\bar{F}Y + X_0^*\bar{F}(Y - Y_0).$$

Use $\|X\|_2 = \|Y\|_2 = 1$ and submultiplicativity. $\square$

6 Five-scale interval audit

The implementation uses 192-bit Arb arithmetic [2]. The algebraic parameter is an interval; midpoint nodes and σ are exact rationals; every exponential, row total, division, absolute difference, row sum, and column sum is outward rounded. Frozen binary64 entries are embedded as their exact dyadic ratios.

Table 1: Validated assembly bounds. The compressed column is the common two-norm upper for coarse and cross-block assembly against the repaired pipeline.

σ	n	max truncation L^1	$\ P - \bar{F}\ _2$	compressed defect	max repair
0.16	32	0	6.377×10^{-16}	8.560×10^{-16}	1.374×10^{-16}
0.08	64	9.088×10^{-17}	1.295×10^{-15}	1.555×10^{-15}	2.528×10^{-16}
0.04	128	8.510×10^{-17}	2.459×10^{-15}	2.767×10^{-15}	3.406×10^{-16}
0.02	256	9.520×10^{-17}	6.175×10^{-15}	6.541×10^{-15}	3.862×10^{-16}
0.01	512	9.577×10^{-17}	1.495×10^{-14}	1.538×10^{-14}	3.776×10^{-16}

The exact Haar embedding defect is

$$\delta_H \leq 8.86511592918 \times 10^{-17}.$$

The smallest repaired pivot lower endpoint over all rows and scales is above 0.0775500, so positivity is separated from zero by fourteen orders of magnitude more than the repair.

The minimum support-center distance from an integer threshold is 9.85×10^{-4} at the finest scale. Thus the archived support pattern is not close to changing under the algebraic parameter enclosure.

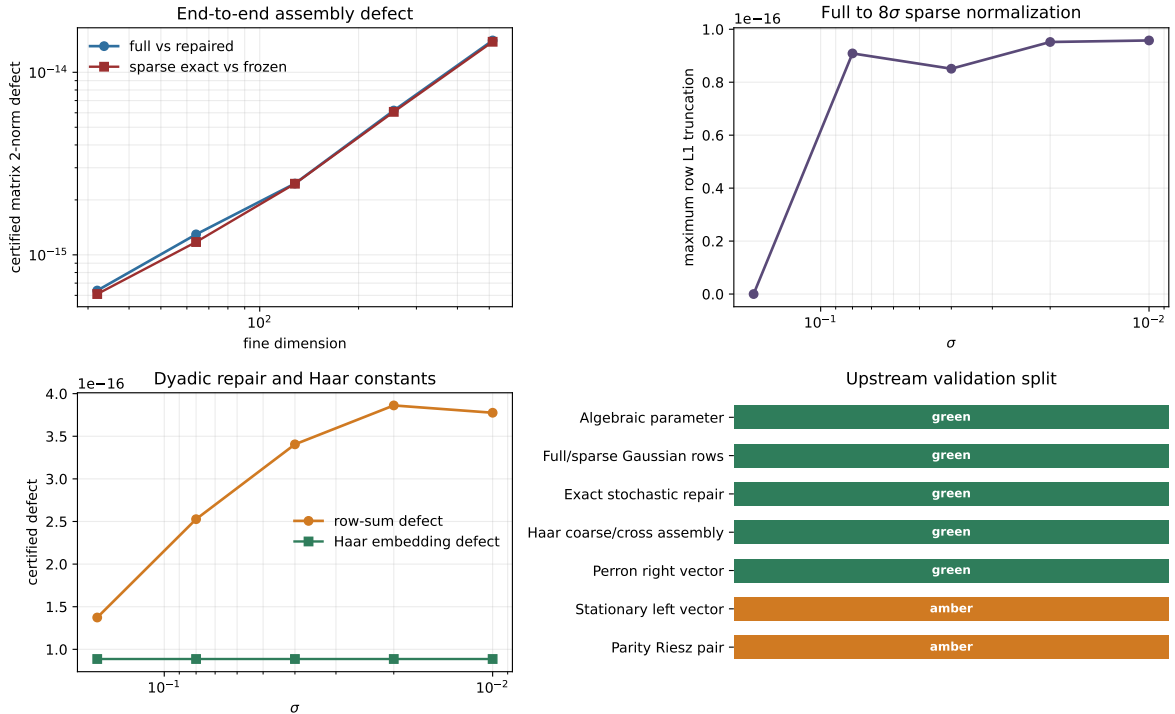
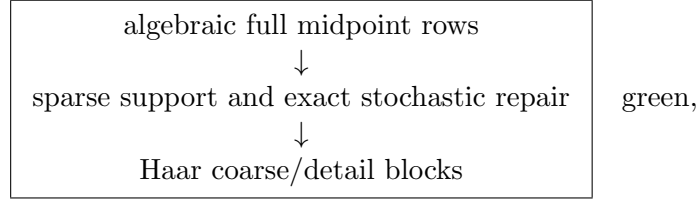


Figure 1: Full/frozen assembly defects, 8σ truncation, exact dyadic repair and Haar constants, and the resulting upstream validation split.

7 Route consequence

The finite-scale upstream graph may now be refined:



while

stationary left factor + parity Riesz pair + rank-two subtraction	amber.
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The exact stochastic repair is especially useful. It removes the need to validate the Perron *right* vector numerically: $\mathbf{1}$ is exact. RH-73 can therefore focus on a stationary left solve and one parity left/right pair, followed by biorthogonal normalization.

8 Boundary and nonclaims

The target here is the fully normalized midpoint matrix, not the continuum operator or exact cell-averaged Ulam matrix. The earlier strong-weak theory provides the analytic midpoint/cutoff route [7]; this paper does not re-prove it.

Five finite scales do not prove a uniform small-noise assembly theorem, although the displayed errors remain at floating-roundoff scale. More importantly, this paper does not validate the stationary left vector, parity eigenvalue/eigenvectors, rank-two projector, deflated bulk operator, or the complete source/observation triple. Hence finite-scale end-to-end Stage A1 is not yet closed.

No Stage A4 identification, pole-renormalized determinant, canonical scattering object, self-adjoint Hilbert–Polya generator, $T \log T$ counting law, prime-power trace formula, completed-zeta identity, or Riemann Hypothesis conclusion is claimed.

9 Conclusion

The folded-Gaussian upstream gate separates cleanly. Positive scalar assembly—including the algebraic parameter, full midpoint weights, 8σ support, row normalization, stochastic repair, and Haar compression—is rigorously stable at all five production scales. The largest complete matrix defect is 1.5×10^{-14} .

The next wall is genuinely spectral rather than numerical assembly: validate the stationary left vector and parity Riesz pair, then propagate their rank-two subtraction through the source and observation blocks.

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